



AGRICULTURAL PRODUCT DISTRIBUTION SYSTEM



Project Overview

Project vision

Digitalizing the end-to-end B2B agricultural distribution framework from central aggregation hubs to regional retail networks (Dealers).

Domain

Supply Chain, warehouse, logistics

Scope

Partner management (suppliers, dealers), order management, inventory, transportation, accounts receivable and payments.

Team size: 2 (1 BA & 1 Dev)

My Responsibilities

Conducted business process and requirement analysis.

Designed AS-IS / TO-BE workflows and BA documentation.

Created user stories, use cases, and wireframes.

Proposed operational improvement solutions.



BUSINESS PROBLEMS

Manual Operations: Relying on Zalo/Phone/Excel leads to data fragmentation and order inaccuracy.

Inventory Blind Spots: Lack of batch/expiration tracking spikes fresh produce spoilage (15-20%).

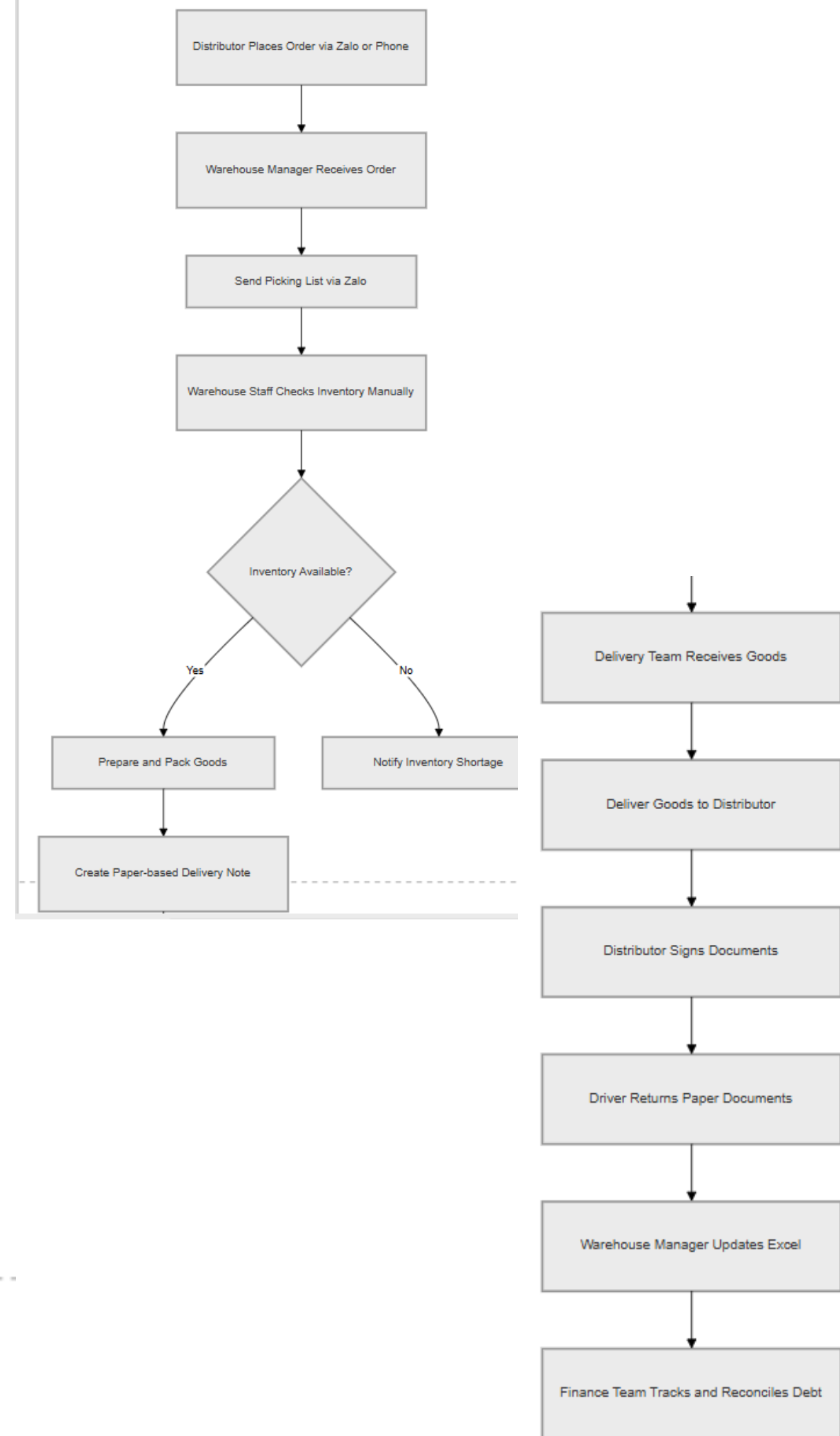
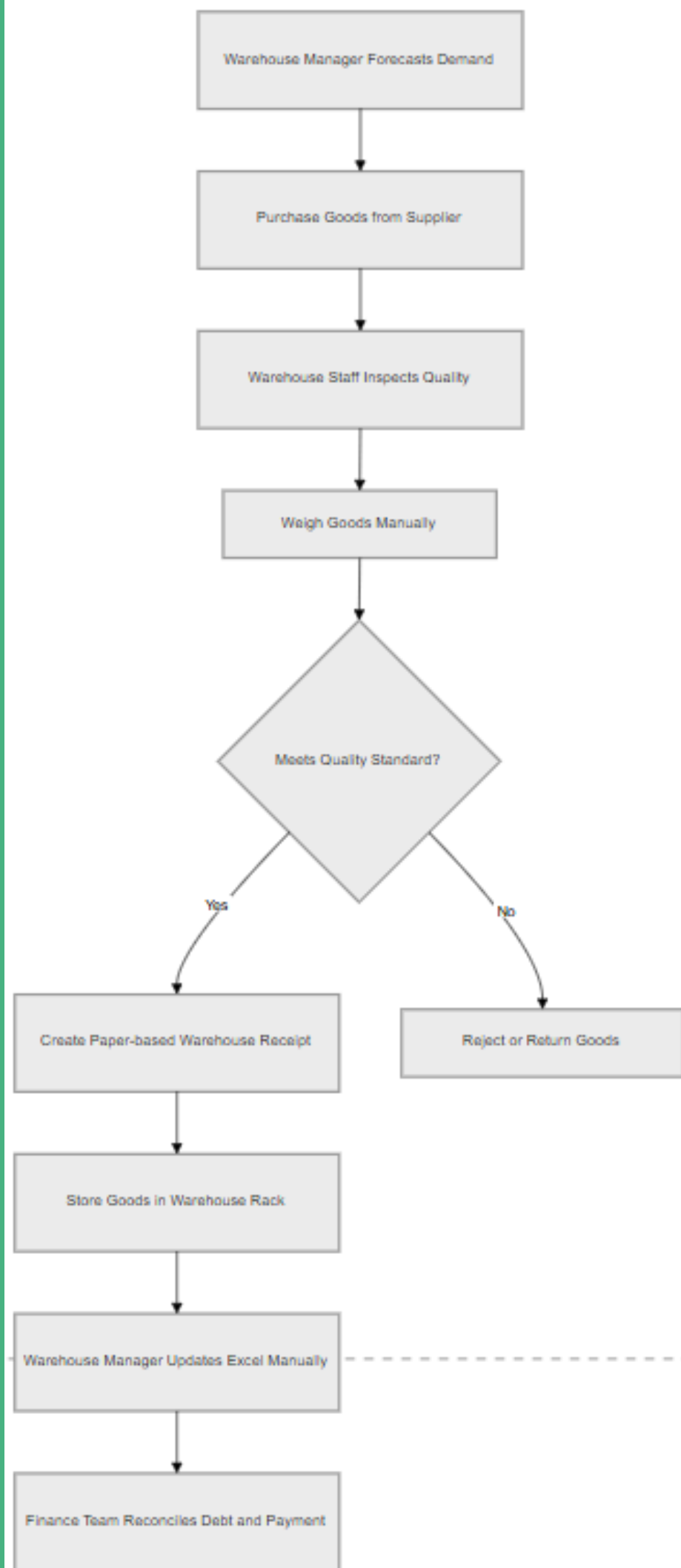
Financial Friction: Delayed paper-based debt reconciliation locks up working capital

Lack of a transportation management system (TMS) and real-time order status update application.



Business Analysis Portfolio Deck

AS-IS WORKFLOW



GAP ANALYSIS

GAP-01

Lack of a centralized data management system.

Very High

GAP-02

Manual inventory management process, leading to a high waste/loss rate.

Very High

GAP-03

Risk of discrepancies in accounts receivable (debt) and financial data.

High

GAP-04

Lack of real-time transportation monitoring capabilities.

Medium

GAP-05

Delayed reporting and a lack of forecasting capability.

Medium

GAP-06

Digital skill barriers among internal staff and external partners.

High

S	W	O	T
• Established Network • Deep Product Knowledge • Well-Defined As-Is Processes	• Legacy Management System • High Inventory Loss Rate • Communication Silos • Poor Scalability	• Digital Transformation Trends • Growing Demand for Transparency • Data-Driven Application	• Fierce Competition • Market Volatility and Natural Disasters • Regulatory Compliance and Food Safety

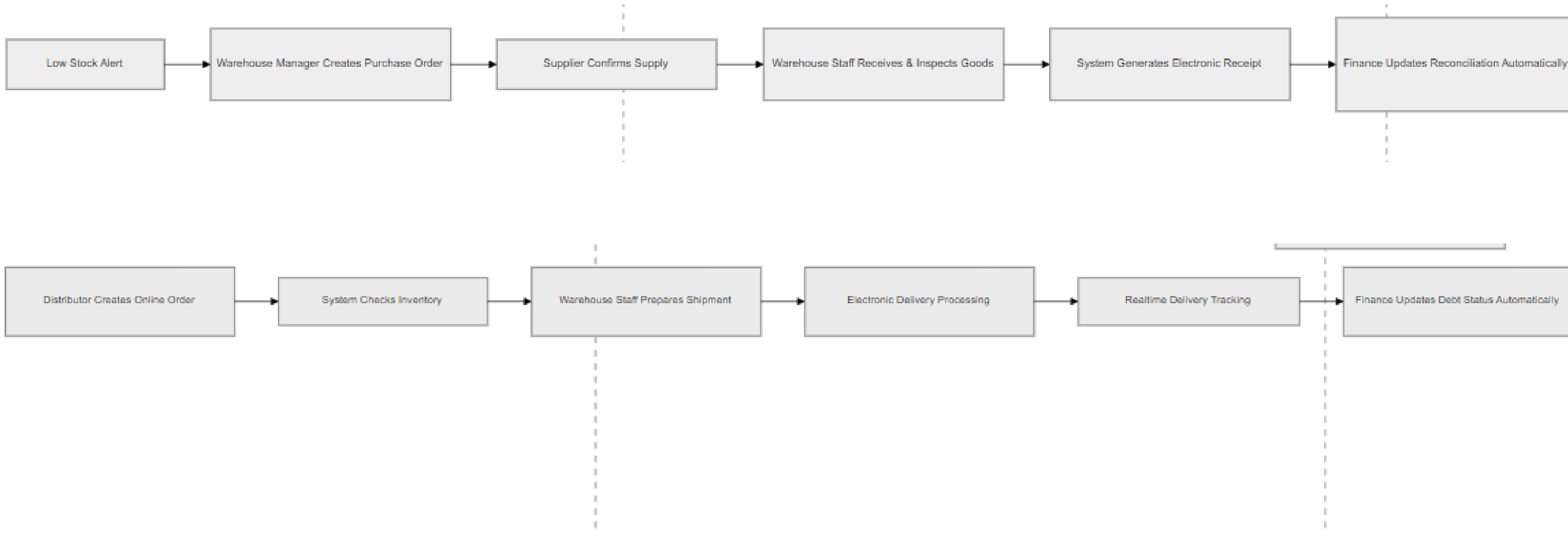
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SUGGEST SOLUTION

Step	Actor	Description
1. Supply & Procurement Management	Supplier	Access the application to update harvest yields, receive purchase orders, and confirm electronic Goods Receipt Notes (e-GRN).
2. Smart Warehouse Management	Warehouse Staff	Scan QR codes for inbound processing; the system automatically assigns storage locations and configures shelf-life alerts (FEFO).
3. Online Ordering	Distributor / Agent	Access the online produce catalog to check available-to-promise (ATP) inventory, place orders, and track real-time discounts directly on the system.
4. Dispatch & Delivery Coordination	Delivery Staff / Logistics Personnel	Receive optimized delivery routes, update order statuses (In Transit / Delivered), and execute electronic proof of delivery (e-POD) with the distributor.
5. Automated Accounts Receivable (AR) Reconciliation	Accountant	The system automatically consolidates outstanding balances based on actual delivery data, dispatches payment reminders, and records instantaneous payment updates.
6. Monitoring & Forecasting	Warehouse Manager	Monitor end-to-end operations via a visual Dashboard, analyze inventory loss/spoilage rates, and forecast procurement demand for the upcoming season.

TO - BE WORKFLOW



FUNCTIONAL SCOPE

Warehouse Operations

Support goods receiving, quality inspection, inventory classification, packing, and shipment preparation

Delivery Management

Coordinate shipment assignments, monitor realtime delivery tracking, and support electronic delivery confirmation

Procurement Management

Create purchase orders, monitor procurement status, approve inbound operations, and coordinate supplier deliveries

Inventory Management

Monitor realtime inventory, track stock movement, generate low-stock alerts, and create electronic warehouse receipts

Order Management

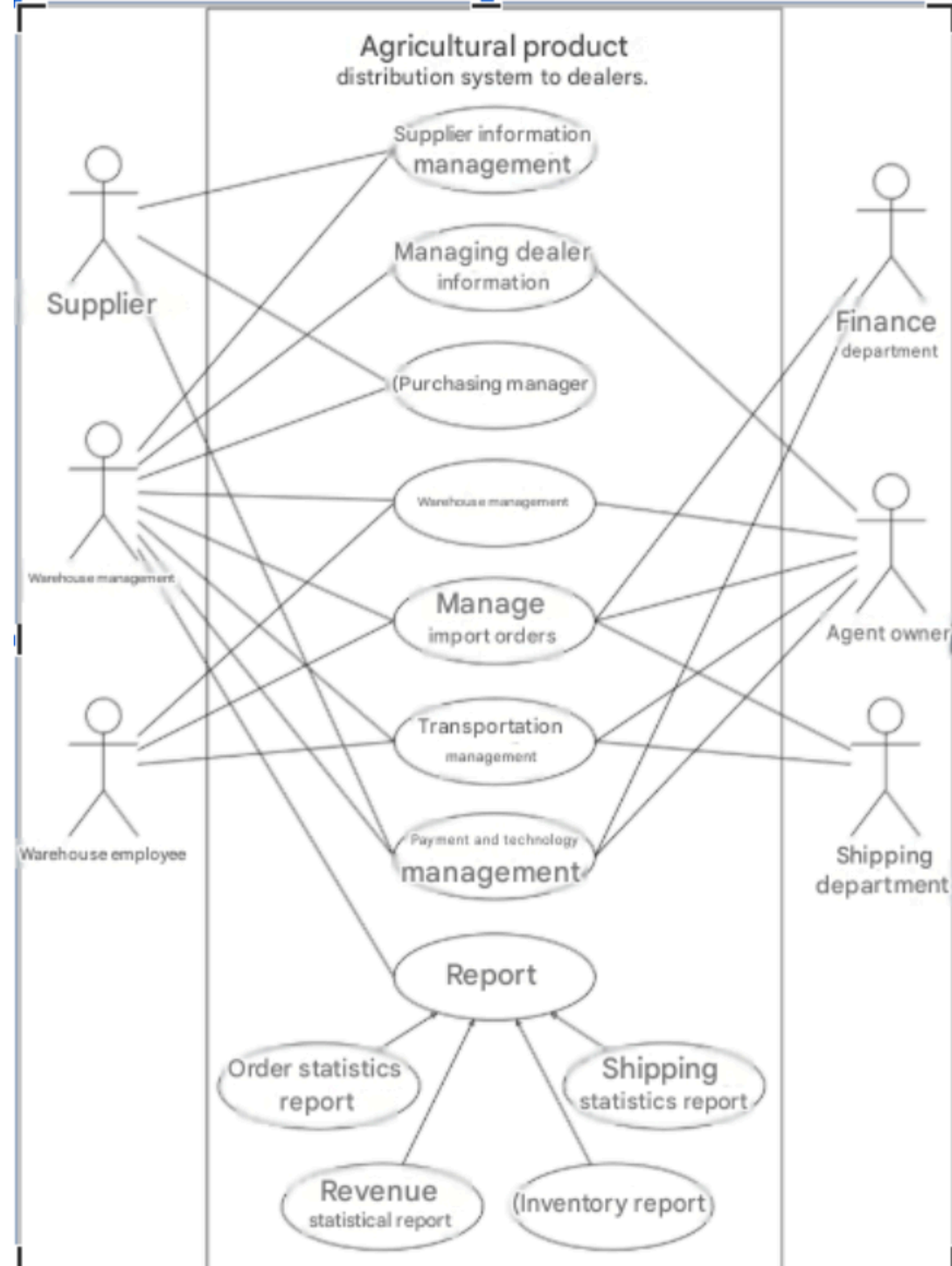
Create distributor orders, verify inventory availability, process order status, and manage order workflows

Payment & Reconciliation

Track supplier and distributor debt, monitor payment status, synchronize financial records, and support automatic reconciliation

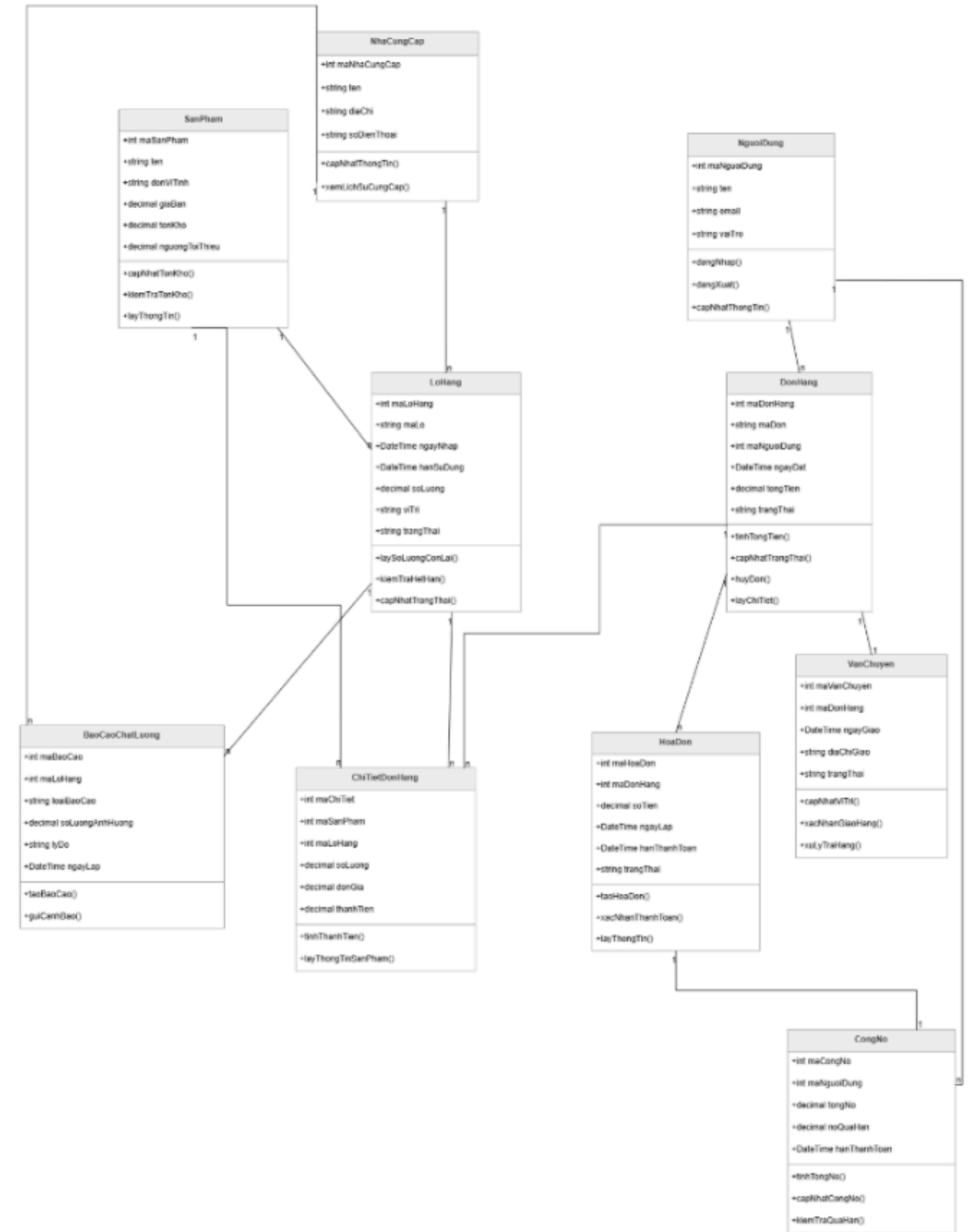
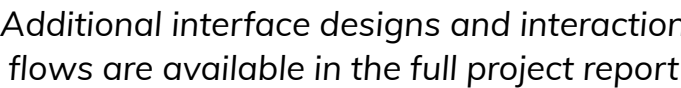
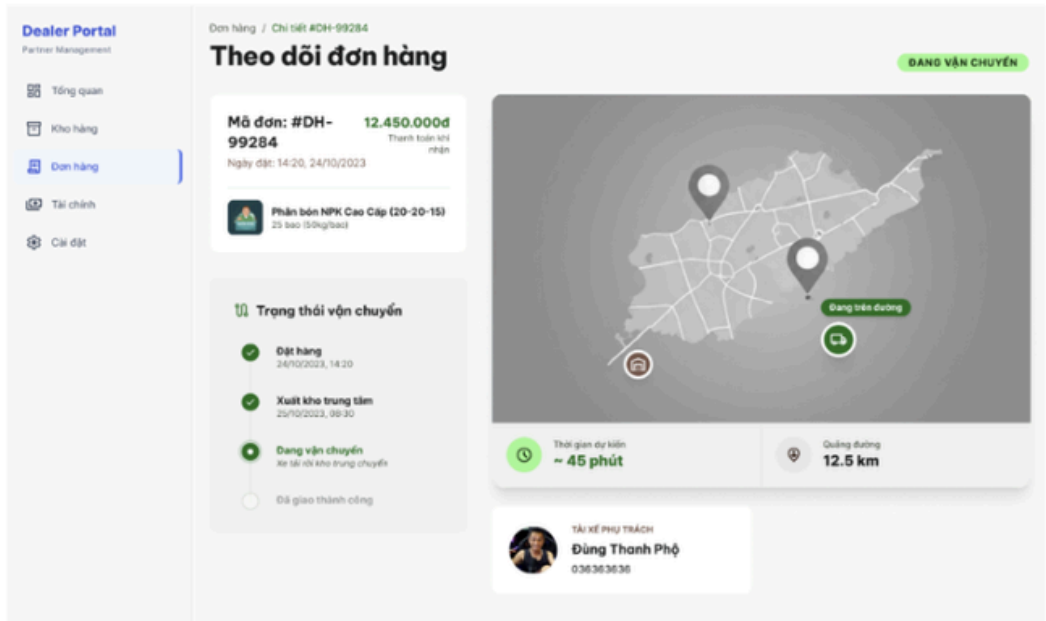
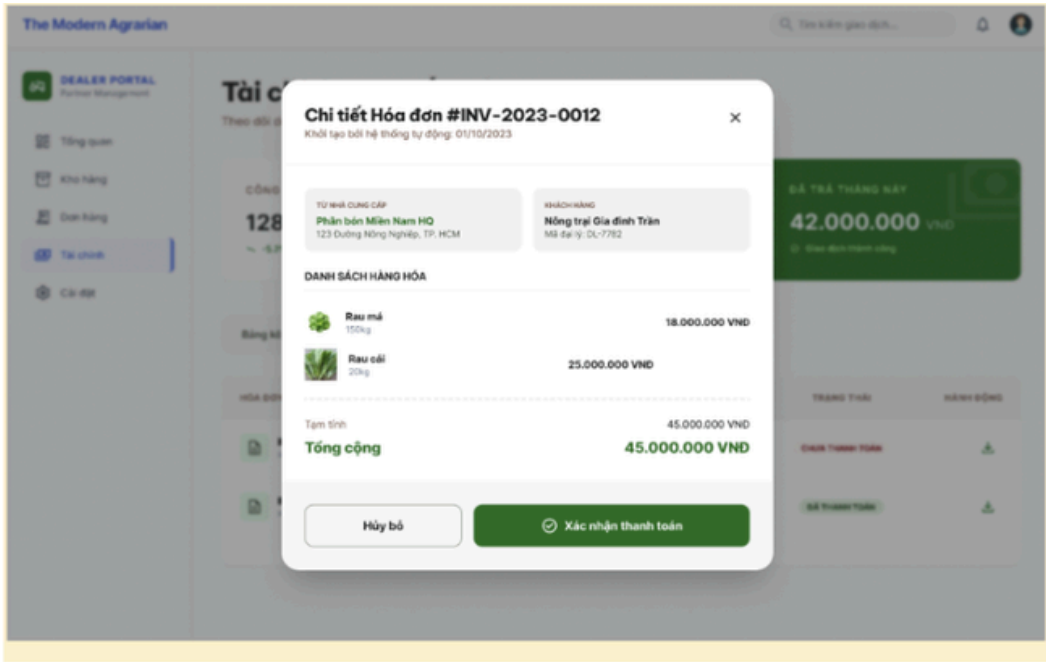


USE CASE DIAGRAM (GENERAL)



```

classDiagram
    class SanPham {
        +int maSanPham
        +string ten
        +string donViTinh
        +decimal giaBan
        +decimal tongKho
        +decimal nguoiTaiThieu
        +capNhatTonKho()
        +kiemTraTonKho()
        +layThongTin()
    }
    class NhaCungCap {
        +int maNhaCungCap
        +string ten
        +string diaChi
        +string soDienThoi
        +capNhatThongTin()
        +xemLichSuCungCap()
    }
    class LoHang {
        +int maLoHang
        +string maLo
        +DateTime ngayNhap
        +DateTime hanSuDung
        +decimal soLuong
        +string viTri
        +string trangThai
        +laySoLuongConLai()
        +kiemTraTaiKhoan()
        +capNhatTrangThai()
    }
    class DonHang {
        +int maDonHang
        +string maDon
        +int maNguoiDung
        +DateTime ngayDat
        +decimal tongTien
        +string trangThai
        +tinhTongTien()
        +capNhatTrangThai()
        +huyDon()
        +layChiTiet()
    }
    class VanChuyen {
        +int maVanChuyen
        +int maDonHang
        +DateTime ngayGiao
        +string diaChiGiao
        +string trangThai
        +capNhatVTC()
        +xacNhanGiaoHang()
        +xulyTraiHang()
    }
    class HoaDon {
        +int maHoaDon
        +int maDonHang
        +decimal soTien
        +DateTime ngayLap
        +DateTime hanThanhToan
        +string trangThai
        +tạoHoaDon()
        +xacNhanThanhToan()
        +layThongTin()
    }
    class CongNo {
        +int maCongNo
        +int maNguoiDung
        +decimal tongHo
        +decimal noQuaHan
        +DateTime hanThanhToan
        +tinhTongNo()
        +capNhatCongNo()
        +kiemTraQuaHan()
    }
    class BaoCao {
        +int maBaoCao
        +int maLoHang
        +string kieuBaoCao
        +decimal soLuong/kinhHuong
        +string lyDo
        +DateTime ngayLap
        +tạoBaoCao()
        +gửiCảnhBao()
    }
    class ChiTietDonHang {
        +int maChiTiet
        +int maSanPham
        +int maLoHang
        +decimal soLuong
        +decimal donGia
        +decimal thanhTien
        +tinhThanhTien()
        +layThongTinSanPham()
    }
    SanPham "1" -- "n" LoHang
    SanPham "1" -- "n" BaoCao
    NhaCungCap "1" -- "n" LoHang
    LoHang "1" -- "n" ChiTietDonHang
    LoHang "1" -- "n" BaoCao
    LoHang "1" -- "n" HoaDon
    LoHang "1" -- "n" DonHang
    LoHang "1" -- "n" VanChuyen
    LoHang "1" -- "n" CongNo
    DonHang "1" -- "n" HoaDon
    DonHang "1" -- "n" VanChuyen
    DonHang "1" -- "n" CongNo
    HoaDon "1" -- "n" CongNo
    
```



Business Value & Operational Risk Reduction

Business
Value

Operational
Risk
Reduction

Reduce manual
workload and
processing time

Improve operational
efficiency through
workflow automation

Increase realtime visibility
across inventory and
delivery operations

Enhance financial
accuracy and
reconciliation processes

Reduce
inventory
mismatch
caused by
manual tracking

Minimize
paperwork
and data
entry errors

Improve
delivery
transparency
and tracking
visibility

Reduce delayed
financial
reconciliation and
communication
gaps

Key Takeaways

- Improved operational efficiency
- Enhanced realtime visibility
- Reduced manual workflow dependency



Thank You for your time and attention